In [136…

```
!pip install pandas numpy seaborn matplotlib scipy --quiet
!pip install nltk --quiet

# Import Libraries
import pandas as pd
import numpy as np
import seaborn as sns
from wordcloud import WordCloud, STOPWORDS
import matplotlib.pyplot as plt
from scipy.stats import pearsonr, spearmanr
import nltk
from nltk.sentiment import SentimentIntensityAnalyzer
nltk.download("vader_lexicon", quiet=True)
sia = SentimentIntensityAnalyzer()
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
```

Step 1: Data Loading and Initial Inspection

The first section of this code will include loading the data and verifying the structure for future use.

In this step:

- Two json files and the three csv files were loaded.
- A dictionary was used to manage the csvs effectively.
- The record counts were printed.
- The first ten rows of all data were then printed to gain an understanding of structure and content.
- This step ensures all data is loaded and visible and some intial information on the dataset is ascertained.

In [137…

```
# Load in the JSON files
products_json_df = pd.read_json("jcpenney_products.json", lines=True)
reviewers_json_df = pd.read_json("jcpenney_reviewers.json", lines=True)

print(f"Products JSON: {len(products_json_df)} rows  |  "
      f"Reviewers JSON: {len(reviewers_json_df)} rows")

# Load in the CSV files
csv_files = {
    name: pd.read_csv(f"{name.lower()}.csv")
    for name in ["Products", "Reviews", "Users"]
}
for name, df in csv_files.items():
    print(f"{name} CSV: {len(df)} rows")
    display(df.head(4))

# Unpack dataframes for later use
products_csv_df, reviews_csv_df, users_csv_df = csv_files.values()
```

```
Products JSON: 7982 rows  |  Reviewers JSON: 5000 rows
Products CSV: 7982 rows
```

| | Uniq_id | SKU | Name | Description | Price | Av_Score |
|---|---|---|---|---|---|---|
| **0** | b6c0b6bea69c722939585baeac73c13d | pp5006380337 | Alfred Dunner® Essential Pull On Capri Pant | Youll return to our Alfred Dunner pull-on capr... | 41.09 | 2.625 |
| **1** | 93e5272c51d8cce02597e3ce67b7ad0a | pp5006380337 | Alfred Dunner® Essential Pull On Capri Pant | Youll return to our Alfred Dunner pull-on capr... | 41.09 | 3.000 |
| **2** | 013e320f2f2ec0cf5b3ff5418d688528 | pp5006380337 | Alfred Dunner® Essential Pull On Capri Pant | Youll return to our Alfred Dunner pull-on capr... | 41.09 | 2.625 |
| **3** | 505e6633d81f2cb7400c0cfa0394c427 | pp5006380337 | Alfred Dunner® Essential Pull On Capri Pant | Youll return to our Alfred Dunner pull-on capr... | 41.09 | 3.500 |

```
Reviews CSV: 39063 rows
```

| | Uniq_id | Username | Score | Review |
|---|---|---|---|---|
| **0** | b6c0b6bea69c722939585baeac73c13d | fsdv4141 | 2 | You never have to worry about the fit...Alfred... |
| **1** | b6c0b6bea69c722939585baeac73c13d | krpz1113 | 1 | Good quality fabric. Perfect fit. Washed very ... |
| **2** | b6c0b6bea69c722939585baeac73c13d | mbmg3241 | 2 | I do not normally wear pants or capris that ha... |
| **3** | b6c0b6bea69c722939585baeac73c13d | zeqg1222 | 0 | I love these capris! They fit true to size and... |

```
Users CSV: 5000 rows
```

| | Username | DOB | State |
|---|---|---|---|
| **0** | bkpn1412 | 31.07.1983 | Oregon |
| **1** | gqjs4414 | 27.07.1998 | Massachusetts |
| **2** | eehe1434 | 08.08.1950 | Idaho |
| **3** | hkxj1334 | 03.08.1969 | Florida |

Step 2: Data Understanding and Quality Check

Before cleaning and merging, the datasets are explored to assess the structure and quality.

In this step:

- Column names are reviewed for each dataset.
- Missing values are identified and summarised.
- A visual heatmap highlights patterns of missing data.
- This step ensures that any inconsistencies or incomplete records can be addressed before further analysis.

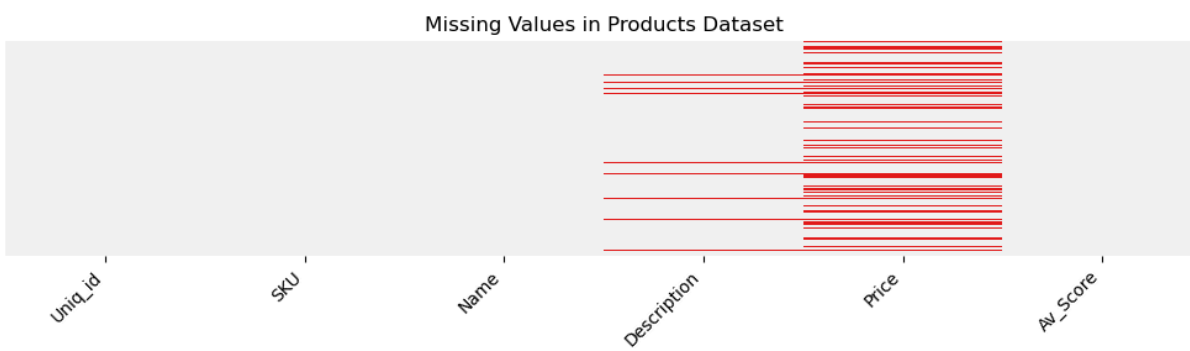
The decision was made to visiualise the data as after it was identified that there were only missing values in one dataset, a heatmap was created to quickly reveal patterns. The heatmap offers an efficient way to compare multiple personas across several behavioural metrics simultaneously. By encoding values using colour intensity, it enables immediate recognition of high and low patterns without relying heavily on numerical scales. Including the real metric values inside each cell maintains analytical precision while keeping the visual uncluttered. This combination preserves clarity and supports quick, meaningful comparison of persona strengths and weaknesses.

In [138…

```
# Step 2: Data Understanding and Quality Check

# Collect all datasets into one dictionary
datasets = {
    "Products CSV": products_csv_df,
    "Reviews CSV": reviews_csv_df,
    "Users CSV": users_csv_df,
    "Products JSON": products_json_df,
    "Reviewers JSON": reviewers_json_df
}
# Create a summary of rows, columns, and total missing values
summary = []
for name, df in datasets.items():
    summary.append([
        name,
        df.shape[0],
        df.shape[1],
        df.isna().sum().sum()
    ])
summary_df = pd.DataFrame(summary, columns=["Dataset", "Rows", "Columns", "Total Mi
```

```
display(summary_df)
# Visualise missing values for the only dataset with missing entries
plt.figure(figsize=(10, 3))
sns.heatmap(
    products_csv_df.isna(),
    cbar=False,
    cmap=["#f0f0f0", "#e41a1c"]
)
plt.title("Missing Values in Products Dataset")
plt.xticks(rotation=45, ha='right')
plt.yticks([])
plt.tight_layout()
plt.show()
```

| | Dataset | Rows | Columns | Total Missing |
|---|---|---|---|---|
| **0** | Products CSV | 7982 | 6 | 2776 |
| **1** | Reviews CSV | 39063 | 4 | 0 |
| **2** | Users CSV | 5000 | 3 | 0 |
| **3** | Products JSON | 7982 | 15 | 0 |
| **4** | Reviewers JSON | 5000 | 4 | 0 |




Step 3: Data Cleaning and Preperation After assessing the data structure and quality, the next step is to clean and standardise it to ensure consistency across files before merging.

In this step:

- All collumn names were standardised by converting to lowercase and key fields renamed.
- All missing values were flagged
- DOB was changed into age groups for more effective analysis
- Removed duplicates
- Prepare the data for merging by ensuring join keys were consistant.
- The approach to missing data was to flag the data rather than delete the data. This method was most appropriate as it meant that crucial data was not deleted and the record count was preserved for analysis. It meant that the missing data could be filtered out when needed. This allowed for the comparison of products with and without

complete information. The deletion would have resulted in 27% of products being removed due to there being over 2,000 missing prices attached.

The decision was made to group raw age values into age groups, this was because the age values were too granular for meaningful segmentation. Furthermore, the retail industry typically targets broad age demographics therefore this is more digestive for the audience of the data. This has also allowed for clearer visualisation and the construction of customer personas.

In [139…

```
# Step 3: Clean & Prepare Data
# Standardise column names
def clean(df):
    df.columns = df.columns.str.strip().str.lower()
    return df

products_csv_df, reviews_csv_df, users_csv_df, products_json_df, reviewers_json_df
    clean(d) for d in [
        products_csv_df, reviews_csv_df, users_csv_df,
        products_json_df, reviewers_json_df
    ]
]
# Unify ID column names
for d in [products_csv_df, reviews_csv_df, users_csv_df,
          products_json_df, reviewers_json_df]:
    d.rename(columns={"uniq_id":"product_id",
                      "username":"user_id"}, inplace=True)
# Clean price fields + flags
dfp = products_json_df
for col in ["list_price", "sale_price"]:
    if col in dfp:
        dfp[col] = (
            dfp[col].astype(str)
            .str.extract(r"(\d+\.?\d*)")[0]
            .astype(float)
        )
dfp["price_missing_flag"] = dfp[["list_price","sale_price"]].isna().any(axis=1).ast
dfp["discount_flag"] = ((dfp["sale_price"] < dfp["list_price"]) &
                        dfp["sale_price"].notna()).astype(int)
dfp["discount_percent"] = np.where(
    dfp["discount_flag"],
    ((dfp["list_price"] - dfp["sale_price"]) /
     dfp["list_price"] * 100).round(2),
    np.nan
)
#

users_csv_df["dob"] = pd.to_datetime(users_csv_df["dob"], errors="coerce", format="
users_csv_df["age"] = ((pd.Timestamp("today") - users_csv_df["dob"]).dt.days / 365.
# Summary + preview
print(
    "Missing price:", dfp["price_missing_flag"].sum(),
    "| Discounts:", dfp["discount_flag"].sum(),
    "| Users missing DOB:", u["dob"].isna().sum()
)
```

```
display(dfp[["product_id","list_price","sale_price",
             "discount_flag","discount_percent",
             "price_missing_flag"]].head())

display(u[["user_id","dob","age","age_group"]].head())
```

```
Missing price: 2166 | Discounts: 5772 | Users missing DOB: 0
```

| | product_id | list_price | sale_price | discount_flag | discount_percent |
|---|---|---|---|---|---|
| **0** | b6c0b6bea69c722939585baeac73c13d | 41.09 | 24.16 | 1 | 41.2 |
| **1** | 93e5272c51d8cce02597e3ce67b7ad0a | 41.09 | 24.16 | 1 | 41.2 |
| **2** | 013e320f2f2ec0cf5b3ff5418d688528 | 41.09 | 24.16 | 1 | 41.2 |
| **3** | 505e6633d81f2cb7400c0cfa0394c427 | 41.09 | 24.16 | 1 | 41.2 |
| **4** | d969a8542122e1331e304b09f81a83f6 | 41.09 | 24.16 | 1 | 41.2 |

| | user_id | dob | age | age_group |
|---|---|---|---|---|
| **0** | bkpn1412 | 1983-07-31 | 42.3 | 40–60 |
| **1** | gqjs4414 | 1998-07-27 | 27.3 | 25–40 |
| **2** | eehe1434 | 1950-08-08 | 75.3 | 60+ |
| **3** | hkxj1334 | 1969-08-03 | 56.3 | 40–60 |
| **4** | jjbd1412 | 2001-07-26 | 24.3 | <25 |

Step 4: Data Verification and Visual Exploration

After ensuring the datasets were fully cleaned, the next step was to verify their structure and begin exploring key patterns in the data.

In this step:

- Column names for all datasets were printed to confirm consistency after cleaning.
- The eight most common brands were identified and their average sale prices visualised.
- Review, product, and user datasets were merged to link customer location with product pricing.
- Total sales values were calculated for each state and the top ten were plotted to highlight key regional markets.

The lollipop chart communicates differences in average sale price using a clean positional scale, which is one of the most accurate visual encodings. Replacing full bars with markers reduces unnecessary ink and keeps attention on the comparison rather than the chart elements themselves. The horizontal orientation improves readability for brand names and

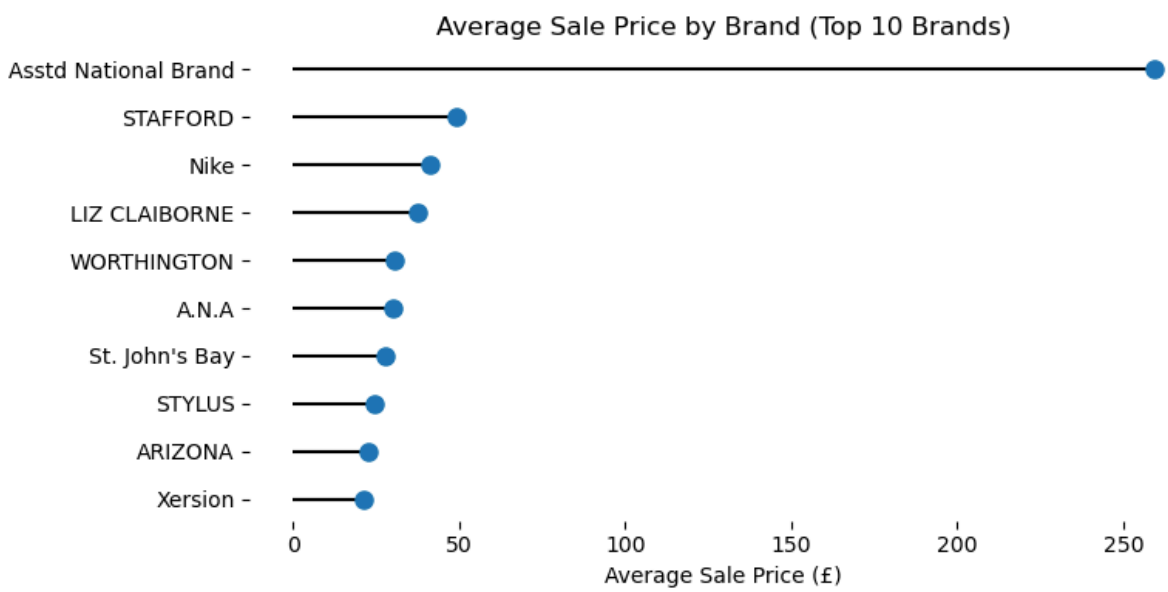
allows for a straightforward ranked layout. Overall, the design highlights price contrasts. The line plot arranges states in descending order, making relative differences in sales value easy to interpret. Connecting the points emphasises the progression across the ranking while keeping the visual simple and minimal. Because the design avoids excessive styling or distracting elements, the viewer can immediately focus on which states contribute the most sales. This approach supports clear, priority-driven insight for decision-making.

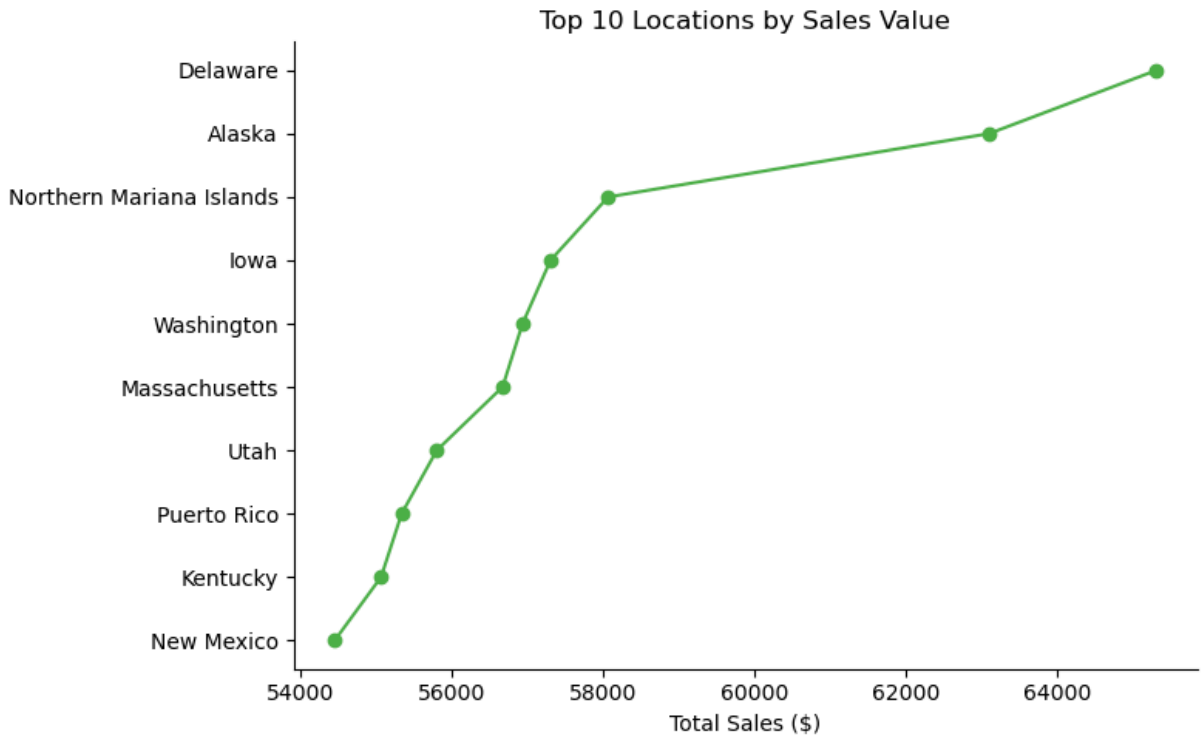
In [140…

```
# Step 1 — Top 10 brands by count
top_brands = (
    products_json_df["brand"]
    .value_counts()
    .head(10)
    .index
)

# Step 2 — Average sale price for those top brands
df_brand = (
    products_json_df[products_json_df["brand"].isin(top_brands)]
    .groupby("brand")["sale_price"]
    .mean()
    .sort_values()
)

# Print columns (optional)
for n, d in {
    "Products": products_csv_df,
    "Reviews": reviews_csv_df,
    "Users": users_csv_df,
    "ProdJSON": products_json_df,
    "RevJSON": reviewers_json_df
}.items():
    print(f"\n{n}:", list(d.columns))

# Step 3 — Lollipop chart: average price by brand
plt.figure(figsize=(8,4))
plt.hlines(y=df_brand.index, xmin=0, xmax=df_brand.values, color="black")
plt.plot(df_brand.values, df_brand.index, "o", markersize=8, color="#1f77b4")

plt.title("Average Sale Price by Brand (Top 10 Brands)")
plt.xlabel("Average Sale Price (£)")
plt.ylabel("")
sns.despine(left=True, bottom=True)
plt.tight_layout()
plt.show()

# Build merged dataset for sales by state
df = (
    reviews_csv_df
    .merge(reviewers_json_df[['user_id','state']], on='user_id', how='inner')
    .merge(products_json_df[['product_id','sale_price','price_missing_flag']],
           on='product_id', how='inner')
    .query("state.notna() & sale_price.notna() & price_missing_flag == 0")
)
```

```
# Calculate total sales per state
sales_by_state = df.groupby("state")["sale_price"].sum().sort_values(ascending=Fals

# Plot top 10 states
vals = sales_by_state.head(10).sort_values()

plt.figure(figsize=(8,5))
plt.plot(vals.values, vals.index, "o-", color="#4daf4a")
plt.title("Top 10 Locations by Sales Value")
plt.xlabel("Total Sales ($)")
sns.despine()
plt.tight_layout()
plt.show()
```

```
Products: ['product_id', 'sku', 'name', 'description', 'price', 'av_score']

Reviews: ['product_id', 'user_id', 'score', 'review']

Users: ['user_id', 'dob', 'state', 'age']

ProdJSON: ['product_id', 'sku', 'name_title', 'description', 'list_price', 'sale_pri
ce', 'category', 'category_tree', 'average_product_rating', 'product_url', 'product_
image_urls', 'brand', 'total_number_reviews', 'reviews', 'bought with', 'price_missi
ng_flag', 'discount_flag', 'discount_percent']

RevJSON: ['user_id', 'dob', 'state', 'reviewed']
```






Step 5: Sentiment Processing

With the review text cleaned and prepared, the next step was to apply sentiment analysis and combine this enriched review data with product and user information to build customer personas.

In this step:

- The VADER sentiment model was loaded and applied to each review to generate a compound sentiment score.
- Review data was further cleaned by standardising column names, removing empty entries, and validating rating values.
- The cleaned review dataset was merged with product details and user demographic data to create a unified view of each customer's interactions.

In [141…

```
# Clean review data
df = (reviews_csv_df.copy()
      .rename(columns={'review':'review_text','score':'rating'})
      .assign(review_text=lambda x: x['review_text'].astype(str).str.strip())
      .dropna(subset=['review_text']))

df['rating'] = pd.to_numeric(df['rating'], errors='coerce')
df = df[df['rating'].between(0, 5)]

# Sentiment scoring
df['sentiment_score'] = df['review_text'].apply(lambda x: sia.polarity_scores(x)['c

print(f"Valid reviews: {len(df):,} | Avg rating: {df['rating'].mean():.2f} | "
```

```
      f"Avg sentiment: {df['sentiment_score'].mean():.2f}")
display(df[['user_id','product_id','rating','sentiment_score','review_text']].head(

# Merge reviews → products → users
merged = (df
          .merge(products_json_df, on="product_id", how="left")
          .merge(users_csv_df, on="user_id", how="left"))

print(f"Merged dataset: {len(merged):,} rows")
```

```
Valid reviews: 39,063 | Avg rating: 1.49 | Avg sentiment: 0.67
```

| | user_id | product_id | rating | sentiment_score | review_text |
|---|---|---|---|---|---|
| **0** | fsdv4141 | b6c0b6bea69c722939585baeac73c13d | 2 | 0.9423 | You never have to worry about the fit...Alfred... |
| **1** | krpz1113 | b6c0b6bea69c722939585baeac73c13d | 1 | 0.8408 | Good quality fabric. Perfect fit. Washed very ... |
| **2** | mbmg3241 | b6c0b6bea69c722939585baeac73c13d | 2 | 0.9514 | I do not normally wear pants or capris that ha... |
| **3** | zeqg1222 | b6c0b6bea69c722939585baeac73c13d | 0 | 0.9329 | I love these capris! They fit true to size and... |

```
Merged dataset: 39,080 rows
```

Step 6: Sentiment Visualisation and Word Analysis

After constructing the persona dataset, visual analysis was used to explore the relationship between sentiment and product ratings, followed by an examination of the most common terms appearing in customer reviews.

In this step:

- A scatterplot was created to show how sentiment scores relate to numerical ratings across users.
- Reviews were grouped into positive, negative, and overall text sets for later exploration.
- Additional stopwords were added to remove generic or uninformative terms commonly found in clothing reviews.
- A wordcloud was generated to highlight the most frequent meaningful words used across all customer reviews.

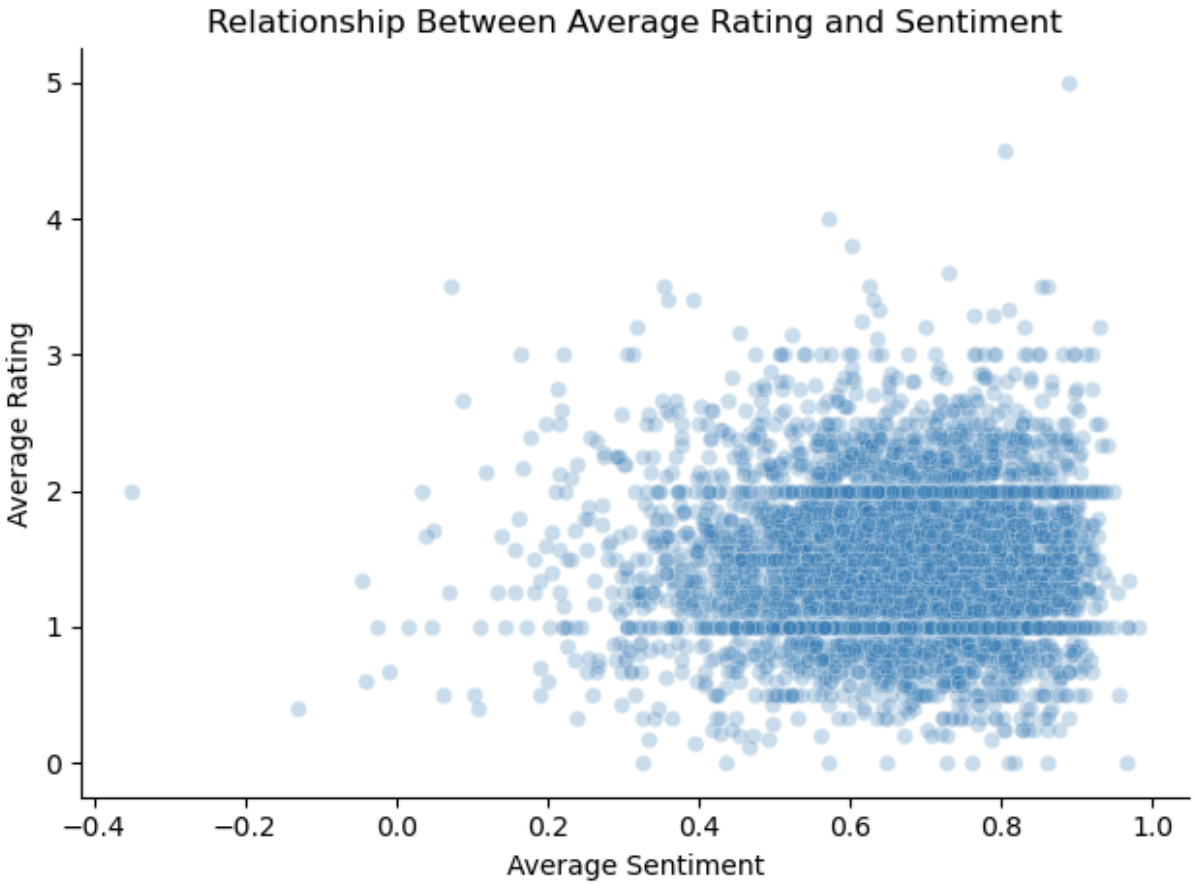
The word cloud offers an impression of dominant topics within customer reviews. It is not intended for precise measurement, however, it is effective for identifying recurring themes or sentiments in unstructured text. Removing common or uninformative words improved the interpretability by highlighting genuinely meaningful vocabulary.

The scatter plot is appropriate here because it directly visualises the relationship between two continuous variables—average sentiment and average rating—using spatial position, which is a precise way to show correlation or lack thereof. Plotting individual customers as points reveals the underlying distribution, outliers, and density patterns without imposing assumptions about the form of the relationship. The light transparency reduces overplotting and keeps the chart readable even with a large number of observations. Overall, this visual effectively communicates whether sentiment scores meaningfully align with user rating behaviour.

In [142…

```
# Build persona_df from merged data
persona_df = (
    merged.groupby("user_id")
    .agg(avg_sentiment=("sentiment_score", "mean"),
         avg_rating=("rating", "mean"))
    .dropna(subset=["avg_sentiment", "avg_rating"])
    .reset_index()
)
# Scatterplot: sentiment vs rating
sns.scatterplot(
    data=persona_df,
    x="avg_sentiment",
    y="avg_rating",
    color="#377eb8",
    alpha=0.25
)
plt.title("Relationship Between Average Rating and Sentiment")
plt.xlabel("Average Sentiment")
plt.ylabel("Average Rating")
sns.despine()
plt.tight_layout()
plt.show()

#  Wordcloud: most common review words
stop = STOPWORDS.union({"dress","buy","bought","size","fit",
                        "product","good","love","nice","color"})
wc = WordCloud(
    width=1000,
    height=600,
    background_color="white",
    stopwords=stop,
    colormap="cool",
    max_words=150
).generate(" ".join(df["review_text"].dropna()))

plt.figure(figsize=(10,6))
plt.imshow(wc)
plt.axis("off")
```

```
plt.title("Most Common Words in All Reviews", fontsize=16, weight="bold")
plt.show()
```



Step 6: Data-Driven Persona Segmentation

To create objective and reliable personas, a K-Means clustering model was applied to customers with complete behavioural data.

In this step:

- Only customers with valid values for rating, sentiment, age, review activity and spend were included, ensuring no imputation or artificial data.
- All features were standardised so each metric contributed equally to the clustering process.
- A stable K-Means model (fixed seeds and multiple initialisations) was used to identify four natural behavioural groups.
- Cluster labels were written back to the full dataset, with incomplete cases left unassigned.
- Cluster centroids were converted back to original units (e.g., actual age and spend) for clear interpretation.

In [143…

```
# Build user-level persona dataset
#Find average of all aspects and number of reviews/products.
#Then calculate favourite category.
persona_df = (
    merged.groupby("user_id")
    .agg({
        "rating": "mean",
        "sentiment_score": "mean",
        "age": "mean",
        "product_id": "count",
        "sale_price": "mean",
        "category": lambda x: x.mode().iloc[0] if not x.mode().empty else "unknown"
    })
    .rename(columns={
        "rating": "avg_rating",
        "sentiment_score": "avg_sentiment",
        "age": "avg_age",
        "product_id": "review_count",
        "sale_price": "avg_price",
        "category": "fav_category"
    })
    .reset_index()
)
# Select clustering features
features = ['avg_rating','avg_sentiment','avg_age','review_count','avg_price']

# Keep only customers with complete data (no imputation)
df_clust = persona_df.dropna(subset=features)
print(f"Using {len(df_clust)} customers for clustering")
print(f"{len(persona_df) - len(df_clust)} customers excluded due to missing values\

# Scale features to equalise influence
```

```python
scaler = StandardScaler()
X_scaled = scaler.fit_transform(df_clust[features])

# Run stable K-means clustering
np.random.seed(42)
kmeans = KMeans(n_clusters=4, random_state=42, n_init=50, max_iter=500)
labels = kmeans.fit_predict(X_scaled)

# Store cluster labels back in the full persona_df
persona_df['cluster'] = pd.NA
persona_df.loc[df_clust.index, 'cluster'] = labels

print("Customers per cluster:")
print(pd.Series(labels).value_counts().sort_index())

# Convert centroids back to original units (for interpretation)
centroids = pd.DataFrame(
    scaler.inverse_transform(kmeans.cluster_centers_),
    columns=features
).round(2)
centroids.index.name = "cluster"

print("\nCluster centroids (original units):")
display(centroids)
```

```
Using 4993 customers for clustering
0 customers excluded due to missing values

Customers per cluster:
0    1869
1     291
2    1848
3     985
Name: count, dtype: int64

Cluster centroids (original units):
```

| cluster | avg_rating | avg_sentiment | avg_age | review_count | avg_price |
|---|---|---|---|---|---|
| **0** | 1.51 | 0.72 | 35.89 | 8.12 | 77.96 |
| **1** | 1.54 | 0.66 | 52.19 | 7.35 | 493.81 |
| **2** | 1.48 | 0.73 | 63.09 | 8.24 | 75.33 |
| **3** | 1.43 | 0.45 | 51.86 | 6.64 | 80.22 |

Step 7: Persona Comparison Visualisation

This step visualises how the four data-driven personas differ across the key behavioural metrics used in the segmentation.

In this step:

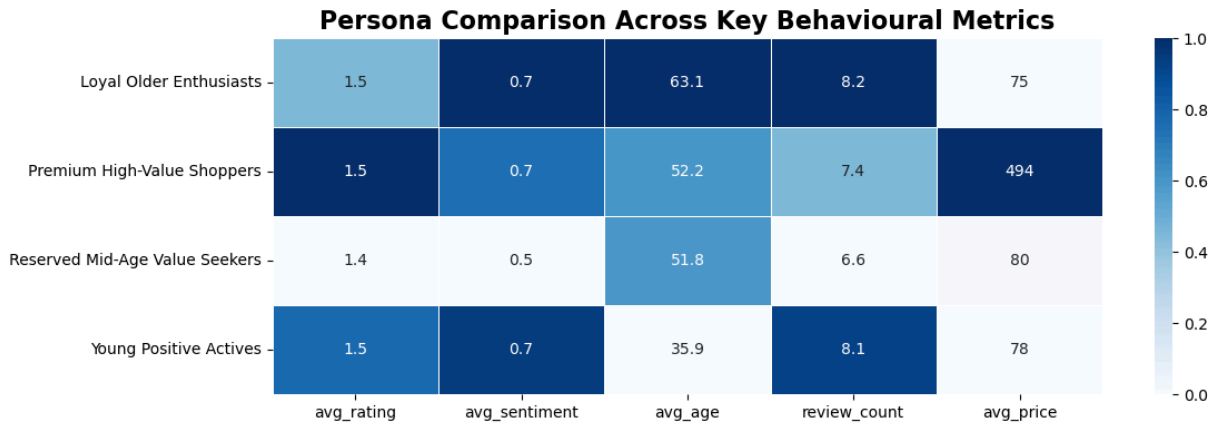
- Final persona labels are applied to each cluster, replacing numeric IDs with meaningful group names.
- Persona-level averages are calculated to capture the typical behaviour of each group.
- These averages are converted into Low / Medium / High bands to place all metrics on a shared scale.
- Minor spacing adjustments prevent overlap while maintaining each metric's band category.
- A comparative line chart is produced to show how each persona positions across the behavioural dimensions.

This heatmap is an effective choice for comparing personas because it condenses multiple behavioural metrics into a single, structured visual layout. Colour intensity immediately highlights relative differences between personas, making patterns in sentiment, spending, engagement, and age easy to spot without scanning multiple charts. Including the actual metric values as annotations preserves numerical accuracy while keeping the design clean and readable.

In [144…

```
# Map cluster IDs to persona names
persona_labels = {
    0: "Young Positive Actives",
    1: "Premium High-Value Shoppers",
    2: "Loyal Older Enthusiasts",
    3: "Reserved Mid-Age Value Seekers"
}
persona_df["persona_name"] = persona_df["cluster"].map(persona_labels)

# Personas, metrics, colours
personas = list(persona_labels.values())
metrics = ["avg_rating","avg_sentiment","avg_age","review_count","avg_price"]
colors = {
    "Young Positive Actives": "#82B2FF",
    "Premium High-Value Shoppers": "#F28E8E",
    "Loyal Older Enthusiasts": "#8FD19E",
    "Reserved Mid-Age Value Seekers": "#FFCE67"
}

# Compute mean metric values per persona
means = (
    persona_df[persona_df["persona_name"].isin(personas)]
    .groupby("persona_name")[metrics]
    .mean()
)

# Normalise for colouring and format labels
norm_means = (means - means.min()) / (means.max() - means.min())
annot = means.copy().round(1)
annot["avg_price"] = means["avg_price"].round(0).astype(int)
annot = annot.astype(str)
```

```
# Heatmap visualisation
plt.figure(figsize=(12,4))
sns.heatmap(norm_means, annot=annot, fmt="", cmap="Blues",
            linewidths=0.4, linecolor="white", cbar=True)

plt.title("Persona Comparison Across Key Behavioural Metrics",
          fontsize=16, weight="bold")
plt.xlabel(""); plt.ylabel("")
plt.tight_layout()
plt.show()
```




In [147…

```
%%writefile custom_style.css
@page {
    size: A4 landscape;
    margin: 1cm;
}

pre, code {
    white-space: pre-wrap !important;
    word-wrap: break-word !important;
    font-size: 8px !important;
}

.container {
    width: 100% !important;
    max-width: 100% !important;
}
```

```
Writing custom_style.css
```

In [146…

```
%%capture
!jupyter nbconvert "2718233_BD2.ipynb" --to webpdf \
  --allow-chromium-download \
  --PDFExporter.latex_count=1 \
  --HTMLExporter.extra_css=custom_style.css
```